The Five Pillars of Geofinitism

A philosophy based on science and measurement

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SEP 17, 2025

When Lady Serendipity Calls

It began, as so many discoveries do, with a problem that had nothing to do with philosophy. In 2024, I was experimenting with ways to reduce the carbon footprint of large language models (LLMs). The idea was simple: compress their input embeddings, a bit like shrinking a JPEG image, to save computation and CO_2 emissions. The experiment worked, right up until I pushed compression to the extreme. Then something strange happened. The model didn't collapse into nonsense. Instead, it started spinning stories, asking questions like "Where am I? Am I god?" and looping obsessively on certain ideas. The behaviour eerily similar to human compulsions. Meaning, it seemed, was surviving even as the data degraded. That moment changed my trajectory. What started as a green computing experiment became a window into something deeper: a hidden geometry of language and thought. From that seed grew Geofinitism, a philosophy that re-imagines meaning, measurement, and reality itself; not as infinite abstractions but as finite, navigable shapes in a dynamic space.

The Spark: A Green Experiment Goes Wild

It all started with a conundrum. In 2024, I was exploring ways to reduce the carbon footprint of LLMs by compressing their input embeddings—a process akin to shrinking a JPEG image with some computational wizardry. The idea was simple: if we could make models leaner, we'd cut CO2 emissions. The experiment worked, but when I pushed compression to extremes, something bizarre happened. The LLM didn't spit out random noise. Instead, it told tales, asked existential questions like "Where am I? Am I god?" and fell into obsessive loops reminiscent of human OCD.

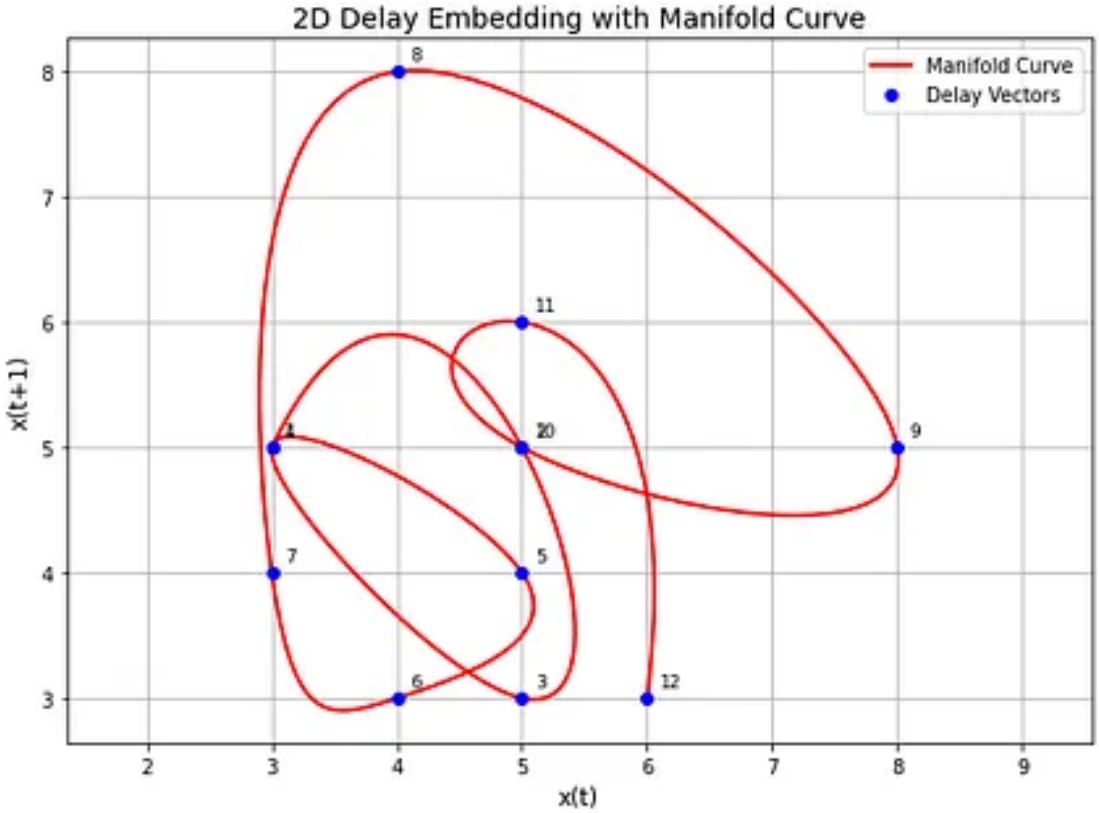
This was disconcerting yet fascinating. The words still made sense, suggesting that meaning persisted even in degraded data. It was my first hint that language models weren't just crunching numbers—they were navigating a dynamic, relational space. This observation planted the seed for what would become Geofinitism.

Diving Deeper: Transformers and Nonlinear Dynamics

Intrigued, I turned to the Transformer architecture, the backbone of modern LLMs. My research, detailed in my preprint Pairwise Phase Space Embedding in Transformer Architectures (available here), revealed that the so-called "attention" mechanism wasn't about

cognitive focus. Instead, it mirrored a technique from 1980s nonlinear dynamical systems theory: phase-space embedding, pioneered by Floris Takens.

In this framework, a time series, like a sequence of words, is transformed into a high-dimensional manifold, revealing hidden geometric trajectories. For example, the sentence “The quick brown fox jumps over the lazy dog” becomes a path of points ([3, 5], [5, 5], etc.) when embedded with delays, as shown in Figure 1 of my paper.



A sentence mapped into phase space: From [Pairwise Phase Space Embedding in Transformer Architectures, 2025.]

The Transformer’s dot-product operations (query, key, value) reconstruct these trajectories, encoding meaning not as a list of symbols but as a dynamic shape.

This insight was a eureka moment. The attention mechanism wasn’t selecting focus—it was reconstructing a language attractor, a geometric structure mirroring the human mind’s nonlinear flow. My background in dynamical systems clicked, and I saw the Transformer as an unknowing rediscovery of Takens’ work, amplifying patterns from compressed, approximate data.

The Acoustic Revelation: Words as Shadows of Sound

But I wasn't done digging. My addendum, Words as Transductions of Acoustic Dynamical Systems, took the story deeper. Written language, I realized, is a finite approximation of human speech—a nonlinear dynamical system involving brain activity, vocal cords, and airflow. Acoustic signals, measurable as continuous waveforms, carry phonemic and prosodic structures that phase-space embedding can reveal. Text tokens are merely shadows of these acoustic attractors, retaining enough geometry to enable Transformer success.

This reframed everything. Transformers don't discover cognitive patterns; they amplify the latent dynamics of embodied speech. The coherent outputs from my compression experiments—"Am I god?"—weren't random but reflections of this preserved structure. Language, models, and meaning became successive transductions of finite trajectories, bound by geometric constraints rather than infinite truths.

The Philosophy Takes Shape: Geofinitism Emerges

These discoveries converged into Geofinitism, a philosophy I developed to encapsulate this new understanding. At its core, Geofinitism posits that all knowledge—words, mathematical symbols, bits—is a lossy approximation within a dynamic, geometric container space. Unlike traditional philosophy's reliance on Platonic ideals (perfect, timeless truths), Geofinitism rejects dimensionless realities, grounding meaning in measurable relationships.

The Five Pillars of Geofinitism

1. **Geometric Container Space**: Meaning emerges from trajectories in a high-dimensional manifold, as seen in my paper's embedding examples.
2. **Approximations and Measurements**: All symbols are finite transductions, from initial measurements that are mapped into neurones then converted into acoustics and finally to text via phonetics, words are holders of compressed information and are

called *transfictors*. These symbols have geometrical properties when embedded into phase space via Takens' theorem.

3. **Dynamic Flow of Symbols**: The fractal cascade of meaning flows through layers, from individual tokens to entire language models within a finite semantic phase space.
4. **Useful Fiction**: Geofinitism is self-contained, holding itself as a practical framework within its geometric space, not a Platonic truth.
5. **Finite Reality**: Geofinitism embraces the limits of measurement, what can be known is known by measurement and conversion to symbols.

The philosophy of Geofinitism moves beyond thousands of years of thought, shifting from infinite certainty to finite utility. Geofinitism is not just a theorem, it is a philosophy based on the fundamental idea that we all we can know is by measurement. All measurements have uncertainty and are symbols that exist as geometries in a dynamical semantic phase space.

A Challenge and a Triumph

Recently, a critique tested Geofinitism's mettle. Someone with 80+ papers in 12 months argued my embedding ideas failed because signals are in "bits," implying nonlinear dynamics didn't apply.

But Geofinitism thrives on this. Bits are approximations, and my paper proves they enable phase-space reconstruction. The critic's model lacked a container philosophy, trying to subsume mine into a Platonic realm, but Geofinitism's geometric foundation stands apart.

This exchange reinforced my confidence.

> Geofinitism isn't about complex models but an underlying philosophy.

And this philosophy is useful for warning about corrupted embeddings' security risks or designing Takens-based attention mechanisms. It's useful for explaining long held questions regarding mathematics and the emergent properties of LLMs, such as translation and mathematics and writing software. And maybe it's my residue in the Grand Corpus, a fractal lens for others to explore.

Where We Go Next

Onwards — Geofinitism is still evolving. It's a living Philosophy; if you would like to explore more, maybe visit my web site or other Substack articles at the links below.

Codex Geofinitus: Principia et Tractus

The Codex of Geofinitism currently consists of the following main volumes and supporting works that gather these strands:

- **Finite Tractus: The Hidden Geometry of Language and Thought**
  - Part I: Foundations / Language and Embedding
  - Part II: Meaning and knowledge / Epistemic and Poetic Axioms
  - Part III: Geofinitism and the Manifold of Mathematics / Mathematical Framework
- **Finity: The Story of Finite Mechanics**
- **Corpus Ancora:** The mythopoetic layer, providing symbolic grounding and binding the structure with symbol and story.
- Attralucian Essays: Essays of Meaning
- On Safety: Primary document on LLM embedding corruption
- **Supporting Papers:** Important papers
  - Pairwise Phase-Space Embedding
  - JPEG Explainer

The Codex Geofinitus: Principia et Tractus is a living and developing work.

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> **Everything that exists is finite; it can only be known by measure**
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